

Project Proposal

Project Title

Testing Surprisal Theory with Language Models: Predicting Human Reading Time Using N-gram, PCFG Model and Neural Language Models

Research Question(s) / Hypotheses

This project investigates whether language model surprisal can predict human reading time during sentence comprehension, a central hypothesis in computational psycholinguistics.

Research questions:

1. Do surprisal values computed from language models correlate with human reading times during sentence processing?
2. How do lexical models (n-gram), structural models (PCFG), and neural language models compare in predicting reading time?
3. How do model-internal representations (e.g., embeddings and attention distributions) relate to observed human processing difficulty?

Hypotheses:

- According to Surprisal Theory, words that are less predictable in context should require more processing effort and therefore have longer reading times.
- Neural language models that capture long-range context should produce surprisal estimates that correlate more strongly with reading time than simple n-gram models.

Methods

1. Language Model Construction

Three types of language models will be used:

- Trigram Language Model
- Probabilistic Context-Free Grammar (PCFG) model for syntactic surprisal
- Pre-trained Neural Language Model (e.g., GPT-2 or similar transformer model)

2. Surprisal Calculation

For each word in the dataset, compute **surprisal**:

$$\text{Surprisal}(w) = -\log P(w | \text{context})$$

This represents the information content of a word given its context.

3. Reading Time Prediction

- Extract **human reading time values** from the dataset.
- Compute surprisal for each word using both language models.
- Analysis will be conducted across participants in a between-subject framework to capture variability in processing behavior.
- Evaluate the relationship between:
 - surprisal
 - human reading time

4. Evaluation

Statistical analyses:

- **Correlation analysis (Pearson/Spearman)** between surprisal and reading time
- **Bayesian regression models** will be used to model the relationship between surprisal and reading time.
 1. Estimation of uncertainty in parameters
 2. More interpretable effect sizes
 3. Comparison across models using posterior distributions

5. Representational and Interpretability Analysis

To better understand how computational models capture linguistic processing:

- **Embedding space analysis** will be used to examine how semantic similarity relates to processing difficulty.
- **Transformer attention patterns** will be analyzed to identify whether regions of high attention correspond to increased surprisal or longer reading times.
- These analyses aim to connect internal model representations with cognitive mechanisms of language comprehension.

Dataset(s)

Primary dataset:

- **Natural Stories Corpus**
It is widely used in psycholinguistic studies of sentence processing and language model evaluation.

Key Research Paper References

1. Levy, R. (2008).
Expectation-based syntactic comprehension.
Cognition, 106(3), 1126–1177.
(Introduces the **Surprisal Theory** of sentence processing.)
2. Demberg, V., & Keller, F. (2008).
Data from eye-tracking corpora as evidence for theories of syntactic processing complexity.
Cognition, 109(2), 193–210.
3. Futrell, R., Gibson, E., & Levy, R. (2020).
Lossy-context surprisal: An information-theoretic model of memory effects in sentence processing.
Cognitive Science, 44(3).
4. Traxler, M. J. (2012). *Introduction to psycholinguistics: Understanding language science.* Wiley-Blackwell. Chapter 10 (“Reading”).

Group Size : 1

Media Link : [Computational Psycholinguistics](#)